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https://doi.org/10.1038/s41415-026-9582-1

Artificial intelligence

AI hallucination risks and mitigation strategies

Evidence-based clinical practice is an essential pillar of healthcare, guiding safer and more informed clinical decisions. In contemporary dentistry, this approach is crucial, as professionals must often choose between different treatment options. Faced with this need, artificial intelligence (AI) tools and Large Language Models (LLMs) are increasingly used to access, summarise, and interpret scientific literature, and can also support clinical decision-making.[1] Although these systems offer unprecedented speed in information retrieval, their incorporation into clinical workflows requires careful examination of their accuracy and reliability.

A critical limitation of current LLMs is the generation of plausible but factually incorrect information, known as 'hallucinations'. These errors are not accidental but statistical: many models are optimised to maximise response fluency, prioritising plausible 'guesses' over admitting uncertainty.[2] This becomes evident when models such as ChatGPT fabricate citations or misinterpret clinical guidelines. In dentistry, where diagnosis and prognostic factors depend on highly specific evidence, relying on free-text consultations ('chats') without verification criteria poses a risk of misinformation, which can lead to errors in decision-making.

It is argued that the dental community should move from simple interactions with chatbots towards prompt engineering and structured AI workflows. Prompt engineering, understood as the strategic design of inputs to guide the model's reasoning, acts as a first line of defence against errors.[3] For clinical applications, improvised and unstructured ('ad-hoc') questioning is often insufficient. Instead, the field requires the adoption of organised structures (such as Chain-of-Thought reasoning) and Retrieval-Augmented Generation (RAG) systems. The RAG architecture is particularly vital for evidence-based practice, as it restricts AI to responding only based on retrieved and verifiable biomedical sources (e.g., PubMed, Embase, Scopus, etc), significantly reducing the hallucination rates compared to standard models.[4]

The future of AI in dentistry likely lies not in generic models but in domain-specific agents or systems. Because precise prompt formulation is essential to mitigate errors, such an approach would allow professionals to focus on clinical reasoning while the system ensures the retrieval of reliable evidence. Importantly, these technologies should be viewed as augmentation tools, not replacements. The ultimate responsibility for critical evaluation and clinical decision-making remains with the professional.

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Generative AI and patient complaints

We write to highlight a recent case that underscores the emerging medicolegal challenges posed by the use of generative AI and large language models (LLMs) in patient complaints and dental care communication.

A patient presented to our practice following completion of a stabilisation phase of care that included multiple extractions and provision of a partial denture. Some months later, an existing anterior bridge failed. Dissatisfied, the patient used an AI engine to draft a detailed, evidence-heavy complaint letter, citing academic resources to support allegations of poor planning and negligence.

On review, the complaint was strikingly persuasive, with well-referenced claims that on first reading suggested deviation from accepted practice. However, in discussion with the patient, a clinician asked them to repeat the exercise but this time provide the AI with fuller background details (a change in prompt): the advanced periodontal condition, symptoms at presentation, and prior stabilisation plan. The AI-generated text then produced a far more balanced interpretation, aligning with the actual clinical context and ultimately resolving the complaint amicably.

This scenario illustrates the ease with which patients can now generate compelling, professional-sounding complaints using AI – even when the inputs are incomplete or lack clinical nuance. The risk is clear: partial histories and selective prompts can produce misleading narratives that seem authoritative, increasing stress and time costs for clinicians and potentially generating medicolegal exposure if such letters are sent to regulators or legal representatives.

A particular concern is that legal claims against dentists are often of relatively modest value. Many are brought on the small claims track by litigants in person, without the input or scrutiny of a claimant lawyer. With the assistance of AI, patients might readily be misled about the merit of a claim, and may generate and issue a convincing-sounding court case within minutes. None of the necessary scrutiny, analysis, and judgement will be applied by lawyer or medicolegal expert until the claim is received by the defendant dentist and/or their defence union. This will inevitably bring unwarranted stress to the practitioner and a further burden on the defence unions, the cost of which will be borne by members.

While AI-assisted health communication has promise, clinicians must recognise its potential to complicate complaint resolution, especially in complex treatment planning cases with staged care and multiple providers. We may need to adopt new strategies to clarify histories, document decision-making in more detail, and educate patients about the limitations of AI-generated health content. This case suggests the importance of early, collaborative discussion to contextualise such outputs and avoid escalation.

It also raises broader questions about the medicolegal frameworks in which such AI-assisted complaints will be evaluated, and the professional duty to ensure our records can withstand scrutiny in a world where patients can generate sophisticated and persuasive – but potentially inaccurate – narratives with a few clicks.

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